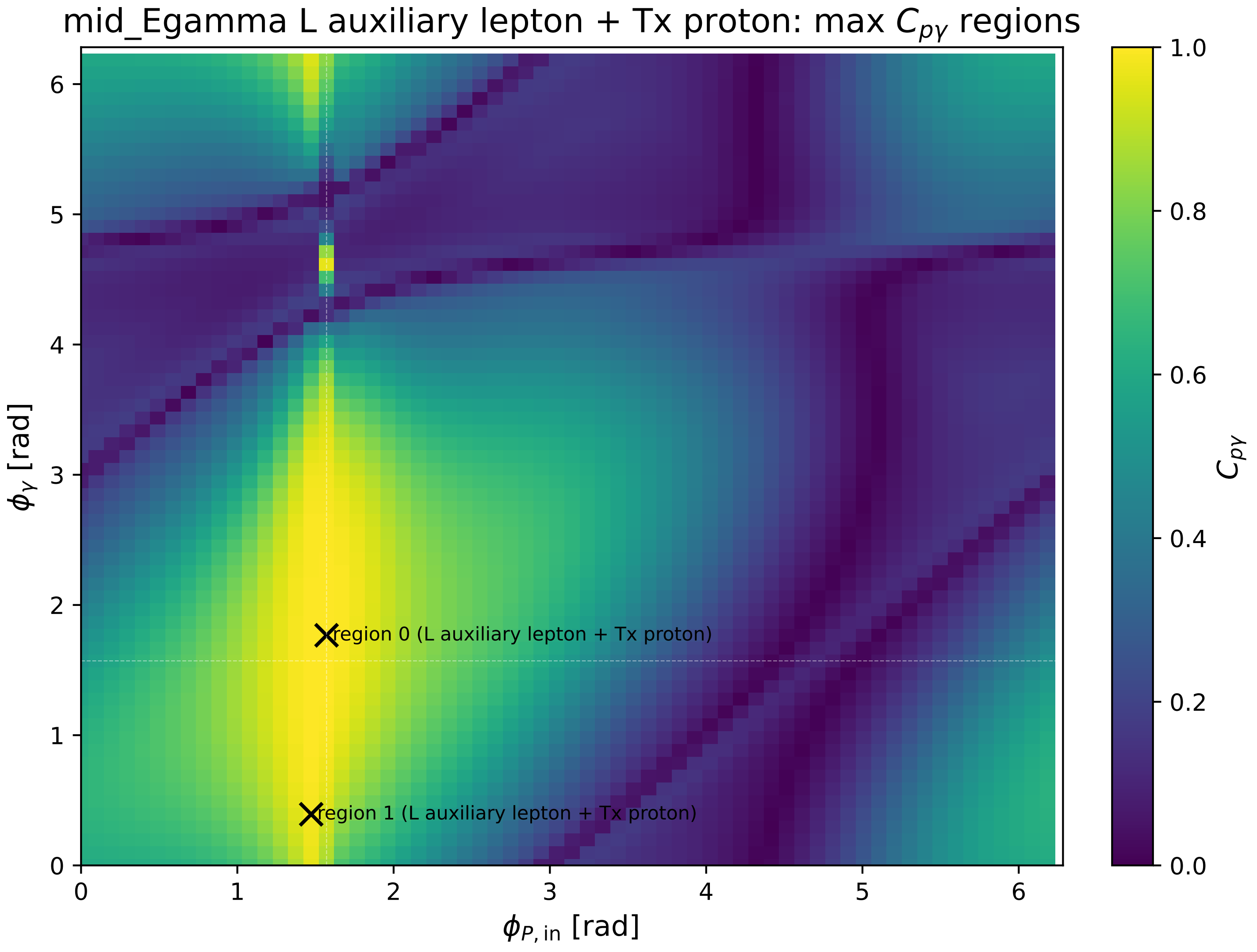
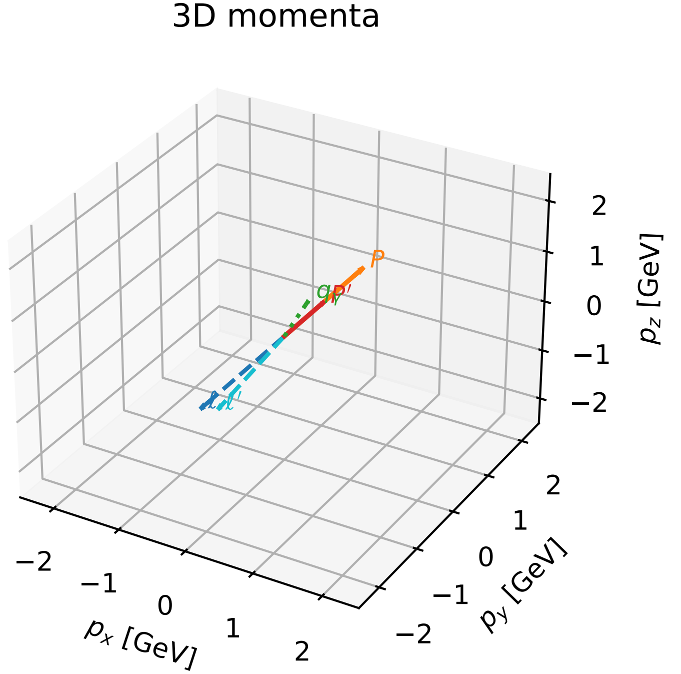

mid_Egamma L auxiliary lepton + Tx proton: max $C_{p\gamma}$ regions
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{p\gamma}$
region 0 (L auxiliary lepton + Tx proton)
region 1 (L auxiliary lepton + Tx proton)
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

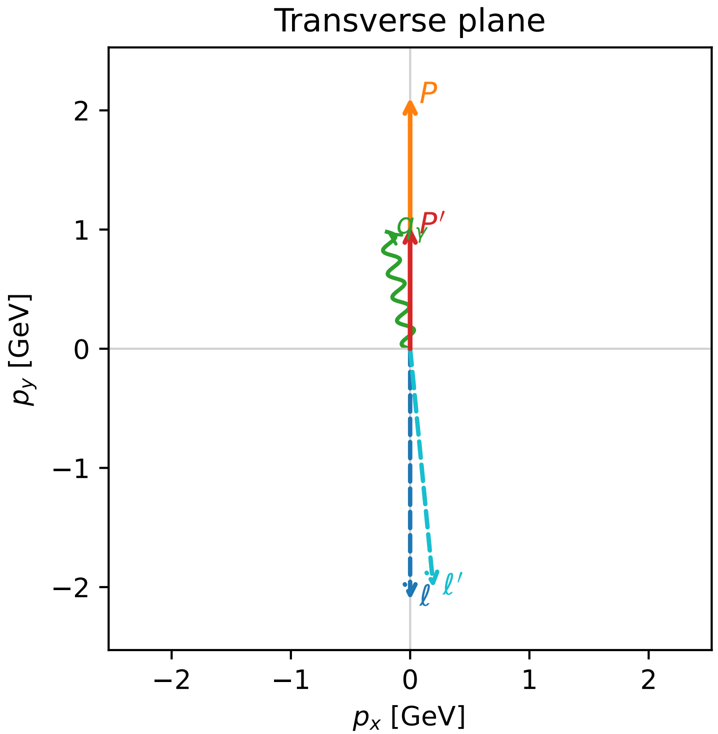
L auxiliary lepton + Tx proton characteristic max $C_{p\gamma}$ configuration

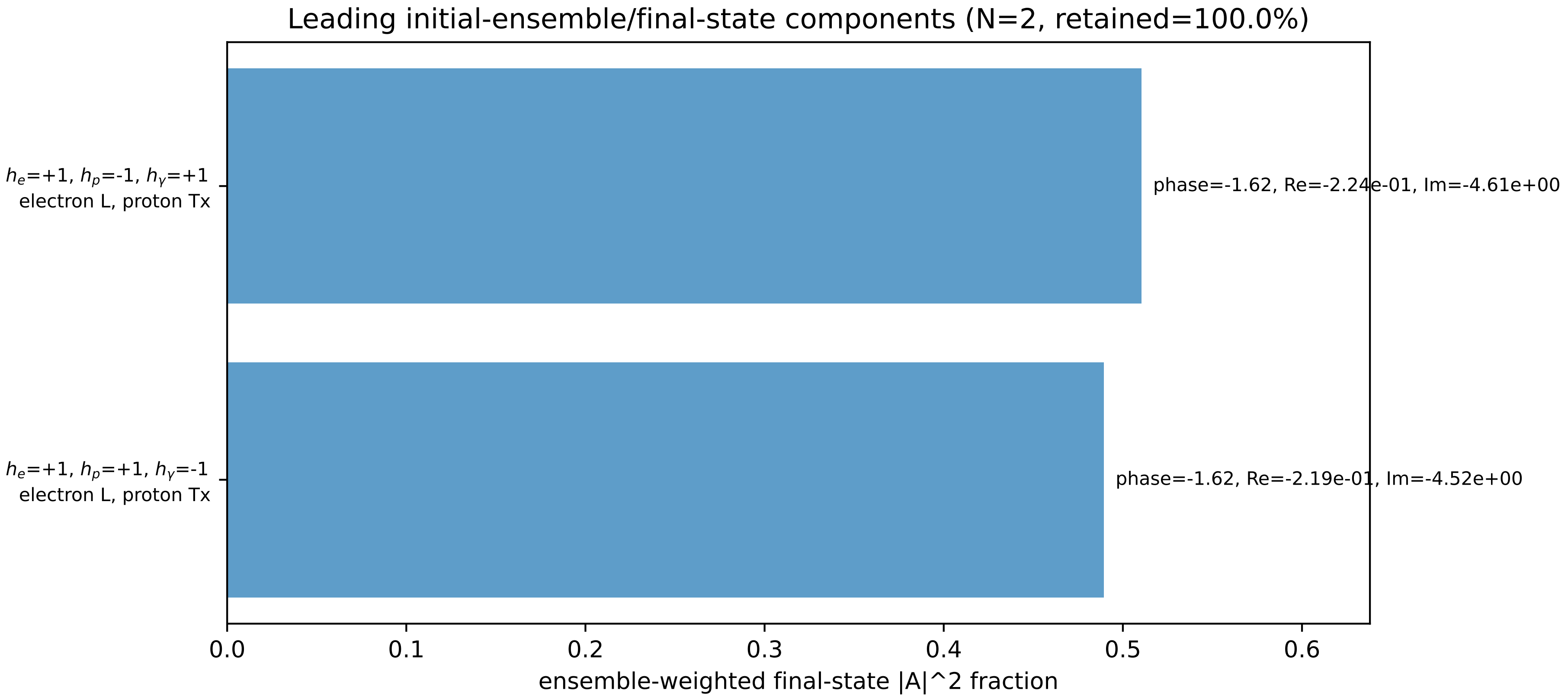


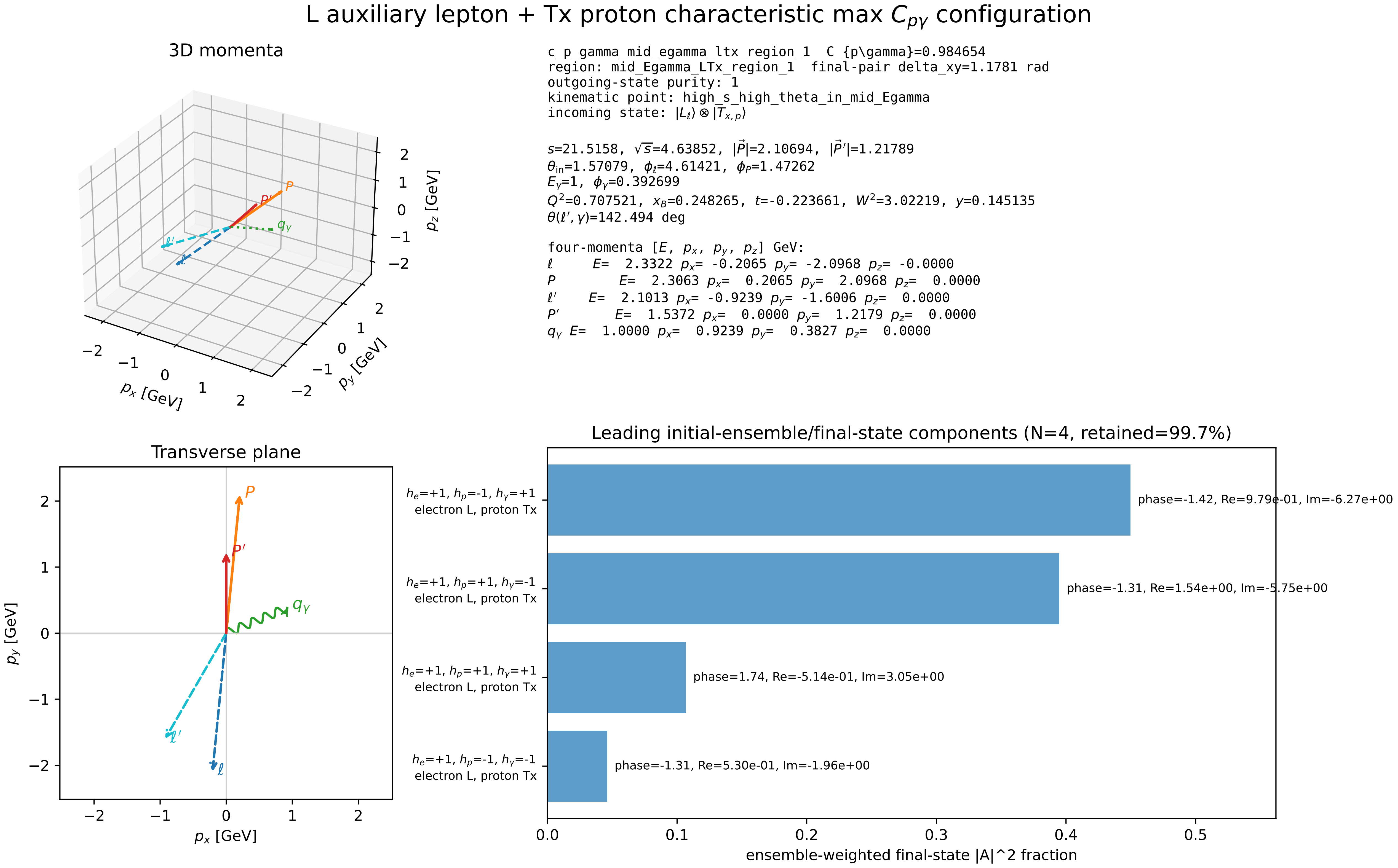

```
c_p_gamma_mid_egamma_ltx_region_0  C_{p\gamma}=0.999659
region: mid_Egamma_LTx_region_0  final-pair delta_xy=0.19635 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
```
incoming state: $|L_\ell\rangle \otimes |T_{x,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.02462
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1, ϕ_γ=1.76715
Q^2=0.041509, x_B=0.0512514, t=-0.330212, W^2=1.64824, y=0.0412461
$\theta(\ell', \gamma)$=174.306 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.0000 p_y= -2.1069 p_z= -0.0000
P E= 2.3063 p_x= 0.0000 p_y= 2.1069 p_z= 0.0000
ℓ' E= 2.2494 p_x= 0.1951 p_y= -2.0054 p_z= 0.0000
P' E= 1.3891 p_x= 0.0000 p_y= 1.0246 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.1951 p_y= 0.9808 p_z= 0.0000







L auxiliary lepton + Tx proton characteristic max $C_{p\gamma}$ configuration
3D momenta
c_p_gamma_mid_egamma_ltx_region_1 C_{p\gamma}=0.984654
region: mid_Egamma_LTx_region_1 final-pair delta_xy=1.1781 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\ell\rangle \otimes |T_{x,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.21789
θ_{in}=1.57079, ϕ_ℓ=4.61421, ϕ_P=1.47262
E_γ=1, ϕ_γ=0.392699
Q^2=0.707521, x_B=0.248265, t=-0.223661, W^2=3.02219, y=0.145135
$\theta(\ell', \gamma)$=142.494 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.2065 p_y= -2.0968 p_z= -0.0000
P E= 2.3063 p_x= 0.2065 p_y= 2.0968 p_z= 0.0000
ℓ' E= 2.1013 p_x= -0.9239 p_y= -1.6006 p_z= 0.0000
P' E= 1.5372 p_x= 0.0000 p_y= 1.2179 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.9239 p_y= 0.3827 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
q_γ
ℓ'
ℓ
Transverse plane
Leading initial-ensemble/final-state components (N=4, retained=99.7%)
h_e=+1, h_p=-1, h_γ=+1 electron L, proton Tx
phase=-1.42, Re=9.79e-01, Im=-6.27e+00
h_e=+1, h_p=+1, h_γ=-1 electron L, proton Tx
phase=-1.31, Re=1.54e+00, Im=-5.75e+00
h_e=+1, h_p=+1, h_γ=+1 electron L, proton Tx
phase=1.74, Re=-5.14e-01, Im=3.05e+00
h_e=+1, h_p=-1, h_γ=-1 electron L, proton Tx
phase=-1.31, Re=5.30e-01, Im=-1.96e+00
ensemble-weighted final-state |A|^2 fraction